

Dynatrace Synthetic Monitoring Guide

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Overview

Dynatrace Synthetic Monitoring helps proactively monitor application availability, performance, and user experience by simulating user interactions from various geographic locations.

Key Benefits

-  Detect issues before end users are impacted
-  Measure response times and availability
-  Test services from multiple locations
-  Validate SLAs and service performance
-  Generate alerts for failures and degradations

Synthetic Monitor Types

Dynatrace supports four primary synthetic monitoring modes:

1.  HTTP Monitor
2.  Browser Monitor
3.  Browser Clickpath Monitor
4.  TCP / ICMP Monitor



1. HTTP Monitor



Purpose

HTTP monitors validate the availability and performance of web services, APIs, and endpoints without rendering a browser.



Common Use Cases

- REST API monitoring
- Health endpoint validation
- Backend service availability checks
- Response time measurement



Configuration Steps

1. Navigate to **Applications & Microservices** → **Synthetic Monitoring**
2. Select **Create Synthetic Monitor**
3. Choose **HTTP Monitor**
4. Enter target URL or API endpoint
5. Configure request method (GET, POST, PUT, etc.)
6. Add headers, authentication, or payload if required
7. Configure execution frequency and locations
8. Save and enable the monitor



Example Screenshot

Insert HTTP Monitor configuration screenshot here.



2. Browser Monitor



Purpose

Browser monitors load complete web pages using a real browser engine and measure end-user experience.



Common Use Cases

- Website availability monitoring
- User experience validation
- Performance baseline measurement
- Third-party dependency monitoring



Configuration Steps

1.  Create a new synthetic monitor
2.  Select **Browser Monitor**
3.  Enter the application URL
4.  Select monitoring locations
5.  Configure execution schedule
6.  Save and activate monitor



Metrics Captured

-  Page Load Time
-  First Contentful Paint
-  Largest Contentful Paint
-  Availability
-  Network Requests



Example Screenshot

Insert Browser Monitor configuration screenshot here.



3. Browser Clickpath Monitor



Purpose

Browser Clickpath monitors simulate complete user journeys across applications.



Common Use Cases

- User login validation
- Checkout flow monitoring
- Search functionality verification
- Multi-step business transaction testing



Configuration Steps

1. Create a synthetic monitor
2. Select **Browser Clickpath**
3. Launch recorder extension
4. Record user actions
 - Click Buttons
 - Enter Credentials
 - Perform Searches
 - Add Items to Cart
5. Validate recorded steps
6. Configure locations and frequency
7. Save and enable monitoring



Best Practices

-  Use dedicated service accounts
-  Avoid hardcoded credentials
-  Include validation checkpoints
-  Keep user journeys optimized

Example Screenshot

Insert Clickpath Recorder screenshot here.

4. TCP / ICMP Monitor

Purpose

TCP and ICMP monitors validate network-level connectivity and service availability.

Common Use Cases

ICMP Monitoring

- Ping validation
- Network reachability checks
- Infrastructure availability verification

TCP Monitoring

- Port availability monitoring
- Database connectivity validation
- Application service verification

Configuration Steps

1.  Create synthetic monitor
2.  Select **TCP Monitor** or **ICMP Monitor**
3.  Enter target hostname or IP address
4.  Configure port number (TCP)
5.  Select execution locations
6.  Define frequency
7.  Save and activate monitor

Example Screenshot

Insert TCP/ICMP monitor configuration screenshot here.

Alerting Configuration

Recommended Alert Settings

-  Availability below threshold
-  Response time degradation
-  Consecutive execution failures
-  Geographic location-specific failures

Integration Options

- Microsoft Teams
- Email
- ServiceNow
- PagerDuty
- OpsGenie
- Webhooks



Monitoring Best Practices

- ✓ Monitor critical business transactions
 - ✓ Use multiple geographic locations
 - ✓ Configure maintenance windows
 - ✓ Create meaningful naming conventions
 - ✓ Review synthetic results regularly
 - ✓ Integrate alerts with incident management tools
 - ✓ Validate monitors after application changes
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Naming Convention Examples

- ◆ HTTP-CustomerPortal-HealthCheck
 - ◆ BROWSER-CustomerPortal-HomePage
 - ◆ CLICKPATH-CustomerPortal-LoginFlow
 - ◆ TCP-Database-Port1433
 - ◆ ICMP-Production-Gateway
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Summary

Dynatrace Synthetic Monitoring provides comprehensive proactive monitoring using:



HTTP Monitoring – API and endpoint validation



Browser Monitoring – Full page performance testing



Browser Clickpath Monitoring – End-to-end user journey validation



TCP/ICMP Monitoring – Network and connectivity verification

Together, these monitoring capabilities provide complete visibility into application, user experience, and infrastructure availability.